Al Imam Mohammad Ibn Saud Islamic University
College of Computer and Information Sciences
Computer Science Department

Course Title:	Design and Analysis of Algorithms
Course Code:	CS216
Course Instructors:	Dr. Fahman Saeed
Exam:	Quiz 1
Semester:	2nd Semester – 2022/2023
Date:	28-12-2022
Duration:	40 minutes
Marks:	7.5 Marks
Privileges:	~~☐ Open Book~~ ~~☐ Open Notes~~ ☐ Calculator Permitted ~~☐ Laptop Permitted~~

Student Name (in English):	
Student ID:	
Section No.:	

Instructions:

1. Answer all questions. There are **2** questions in 2 pages.
2. Write your name on each page of the exam paper.
3. Write your answers directly on the question sheets. Use the ends of the question pages for rough work or if you need extra space for your answer.
4. If information appears to be missing from a question, make a reasonable assumption, state your assumption, and proceed.

Official Use Only		
Question	**Student Marks**	**Question Marks**
1		2.5
2		2
3		3
Total		**7.5**

Student Name (in English): _____ **Student ID:** _____

Question 1: **Multiple choices** To be answered in **(24) Minutes** [ ] / 2.5 **Marks**

Choose one correct answer for each statement and fill the following table:

Question	1	2	3	4	5
Answer	a	c	c	a	d

1) **What is the closet time complexity of the function *f(n) = n log 4 + log n*:**
   - **a.** *O (n)*
   - **b.** *O (n log n)*
   - **c.** *O (log n)*
   - **d.** *O (n log (n^2))*

2) **If *f(n)=sin(n)* and *g(n)=n sin(n)*:**
   - **a.** *f(n)* is *O(g(n))*
   - **b.** *f (n)* is *Ω(g(n))*
   - **c.** Neither *f(n)* is *O(g(n))* nor *f(n)* is *Ω(g(n))*
   - **d.** None of the above

3) **If *f (n) = O (g(n))* and *g(n) = O(h(n))* then:**
   - **a.** *f (n) = Θ (h(n))*
   - **b.** *f (n) = Ω (g((n))*
   - **c.** *h(n) = Ω (f(n))*
   - **d.** *h(n) = O(f(n))*

4) **If *f (n) = O (g(n))* then *f(n)+g(n)* is:**
   - **a.** *O(g(n))*
   - **b.** *O (f(n))*
   - **c.** *O (f(n)*g(n))*
   - **d.** Cannot be determined

5) **If *f(n) = log n + 1*, then:**
   - **a.** *f(n) = Ω (1)*
   - **b.** *f(n) = Ω (log n+1)*
   - **c.** *f(n) = Ω (log n)*
   - **d.** All the above

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Question 2: Asymptotic Notation To be answered in **(15) Minutes [ ] / 2 Marks**

What is the time complexity of the following algorithm in the most precise *Big O*?

A	**B (bonus)**	C
i ←1 for i<= n do j ←3 for j<=i do print j endFor endFor	$\sum_{i=a}^{b} i$ =	log ($24n^5$) **show the steps. Find c and n0?**
Answer: $O(n^2)$	Answer: b(b+1)/2 – a(a+1)/2	Answer: O(logn) c=6 and n0=32

Question 3: Compute To be answered in**(15) Minutes [ ] / 3 Marks**

a) Find the largest input size n that we can get it within **a Week** if we have an algorithm with the running time n^2 and a computer with the performance 5^2 operations per second. **(1 points)**

Total operations in a day = 5^2 * 60 * 60 * 24 * 7
= 15,120,000 operations

The largest input size: n^2 = 15,120,000
n = 3888.44
=> 3889

b) Find the asymptotic relationship between $f(n) = n^{\log n}$ and $g(n) = n^{\sqrt{n}}$ (using ***limit***) **(2 points)**

$\lim_{n\to\infty} \frac{n^{\log n}}{n^{\sqrt{n}}} = \frac{\infty}{\infty}$ Add log

$\lim_{n\to\infty} \frac{\log n^{\log n}}{\log n^{\sqrt{n}}} = \lim_{n\to\infty} \frac{\log n \log n}{\sqrt{n} \log n}$

$= \lim_{n\to\infty} \frac{\log n}{\sqrt{n}} = \frac{\infty}{\infty}$ Use the Lopital theorem

$\lim_{n\to\infty} \frac{\frac{1}{n .\ln 2}}{\frac{1}{2\sqrt{n}}} = \lim_{n\to\infty} \frac{2\sqrt{n}}{n .\ln 2}$ Multiply by $\sqrt{n}$

$= \lim_{n\to\infty} \frac{2}{\sqrt{n} .\ln 2} = \frac{2}{\infty} = 0$

f(n) = O g(n)